

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I D. Dhanshana (1925²⁴²⁴³) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

D. Dhanshana
Signature of the student:

20+27+20+18+9
(94)
bp

Student 9: Dharshana D | Register Number: 192524243

1. Question 25 - Search Strategy Comparison

Consider a graph with $S \rightarrow A, B$; $A \rightarrow C, D$; $B \rightarrow E$; $C \rightarrow G$; $D \rightarrow G$; $E \rightarrow G$.

Perform BFS, DFS, UCS, Greedy Best-First Search and A* separately. Complete the comparison table.

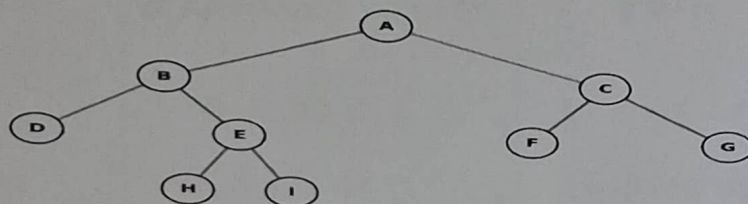
Algorithm	Expansion Order	Path	Cost	Complete?	Optimal?
1					
2					
3					
4					
5					

2. Question 1 - Breadth-First Search (BFS)

Consider the following graph.

Task: Starting from node A, perform Breadth-First Search (BFS).

Complete the table below.

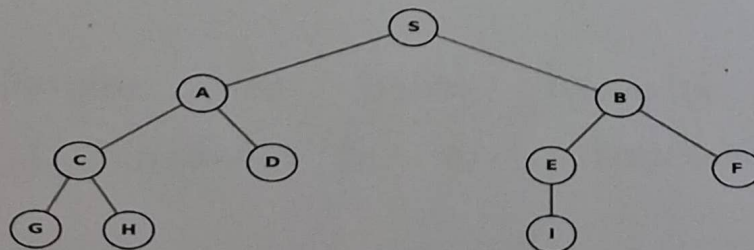


Iteration	Queue Before	Expanded	Queue After
1			
2			
3			
4			
5			

3. Question 2 - Depth-First Search (DFS)

Using the following graph, perform Depth-First Search (DFS) starting from node S.

Complete the table below.



Iteration	Stack Before	Expanded	Stack After
1			
2			
3			
4			
5			

Search Strategy Comparison

Given:

$S \rightarrow A, B$

$A \rightarrow C, D$

$B \rightarrow E$

$C \rightarrow G$

$D \rightarrow G$

$E \rightarrow G$

Assuming each edge has cost 1 and the goal is G :

Algorithm	Expansion Order	Path	Cost	Complete	Optimal
BFS	S, A, B, C, D, E, G	$S \rightarrow A \rightarrow C \rightarrow G$	3	Yes	Yes
DFS	S, A, C, G	$S \rightarrow A \rightarrow C \rightarrow G$	3	Yes	Yes
UCS	S, A, B, C, D, E, G	$S \rightarrow A \rightarrow C \rightarrow G$	3	Yes	Yes
Greedy Best-First	S, A, C, G	$S \rightarrow A \rightarrow C \rightarrow G$	3	Yes	No
A^*	S, A, C, G	$S \rightarrow A \rightarrow C \rightarrow G$	3	Yes	Yes

Assume the usual heuristic that makes c preferred toward G as heuristic values.

2. Breadth - First Search (BFS)

Starting Node: A

Graph:

- $A \rightarrow B, C$
- $B \rightarrow D, E$
- $C \rightarrow F, G$
- $D \rightarrow H, I$

Iteration	Queue Before	Expanded	Queue After
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G, H, I

BFS Traversal:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I$

If the search is supposed to stop when G is found, then the traversal stops at G.

3. DFS - Depth - First Search

Starting Node: S

Iteration	Stack Before	Expanded	Stack After
1	S	S	B, A
2	B, A	A	B, D, C
3	B, D, C	C	B, D, H, G
4	B, D, H, G	G	B, D, H
5	B, D, H	H	B, D
6	B, D	D	B
7	B	B	F, E
8	F, E	E	F, I
9	F, I	I	F
10	F	F	-

DFS Traversal:

$S \rightarrow A \rightarrow C \rightarrow G \rightarrow H \rightarrow D \rightarrow B \rightarrow E \rightarrow I \rightarrow F$

Since, DFS uses a stack (LIFO), children are pushed in reverse order so that the left child is expanded first.